

```
{
    Container c = getContentPane();
    c.setLayout( new FlowLayout() );
    numLabel = new JLabel( "Enter and Integer and press
Enter" );
    c.add( numLabel );
    num = new JTextField( 10 );
    // creates a new JTextField with 10 columns
    num.addActionListener( this );
    // makes "this" Applet listen for events.
    c.add( num );
    resultLabel = new JLabel( "Fibonacci value is " );
    c.add( resultLabel );
    result = new JTextField( 15 );
    result.setEditable( false );
    c.add( result );
} // end of method init()
public void actionPerformed( ActionEvent e )
{
    long number,
    fibonacciValue;
    number = Long.parseLong( num.getText() );
    showStatus( "Calculating..." ); // status area of Applet
    fibonacciValue = fibonacci( number );
    showStatus( "Done" );
    result.setText( Long.toString( fibonacciValue ) );
} // end of method actionPerformed()
// Recursive definition of method Fibonacci
public long fibonacci( long n )
{
    if( n == 0 || n == 1 ) // base case
    {
        return n;
    }
    else
    {
        return fibonacci( n - 1 ) + fibonacci( n - 2 );
    }
}
```